

DIABETES MELLITUS

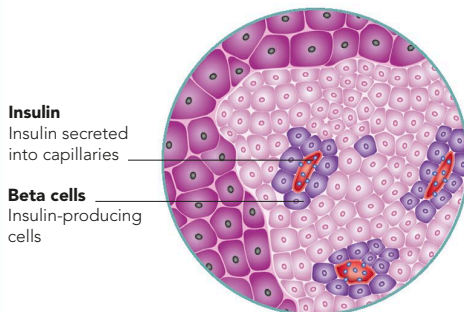
THE MAIN ENERGY SOURCE FOR CELLS IS GLUCOSE, WHICH THE CELLS ABSORB FROM THE BLOOD WITH THE HELP OF THE HORMONE INSULIN. IN DIABETES MELLITUS, THIS PROCESS DOES NOT WORK PROPERLY. THERE ARE TWO MAIN TYPES OF DIABETES MELLITUS: TYPE 1 AND TYPE 2.

TYPE 1 DIABETES

Type 1 diabetes mellitus is an autoimmune disorder. It occurs when the immune system misidentifies beta cells in the islets of Langerhans in the pancreas as foreign and destroys them.

The cause is unknown, but the disease may be triggered by a viral infection or inflammation in the pancreas. It usually develops in childhood or adolescence. Symptoms include thirst, dry mouth,

hunger, frequent urination, fatigue, blurred vision, and weight loss. If untreated, the disorder can cause ketoacidosis, in which toxic chemicals called ketones build up in the blood. Affected people need urgent medical attention; otherwise they can fall into a coma. There can also be long-term complications (see Type 2 diabetes, opposite). Treatment involves insulin injections.

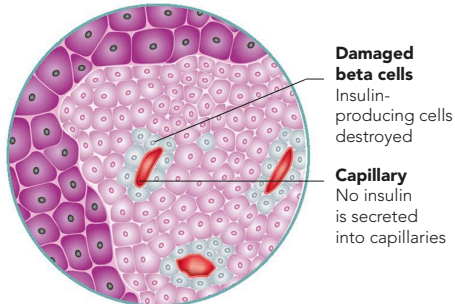


DAMAGED BETA CELLS

If the beta cells are damaged, they cannot produce insulin. As a result, body cells cannot take up glucose, and blood glucose levels rise too high. The lack of insulin allows the alpha cells to produce more glucagon, which raises blood glucose levels still further.

NORMAL BETA-CELL FUNCTION

As food and drink are digested, the presence of glucose, amino acids, and fatty acids in the intestine stimulates beta cells to release insulin into the bloodstream via tiny blood vessels called capillaries, which run through the islets of Langerhans.



BLOOD SUGAR REGULATION

During digestion food is broken down to provide substances that cells can use to fuel and repair themselves. The main source of fuel is glucose, which is carried in the bloodstream to cells. Any excess is stored in the liver, muscle cells, and fat cells. The body needs to keep the blood glucose level steady. If it is too low, cells will not have

enough glucose for their energy needs. If it is too high, there is a risk of autoimmune disease and pancreatitis. If the blood glucose level is too low, alpha cells in the islets of Langerhans secrete glucagon, which stimulates the release of stored glucose. If the level is too high, beta cells in the islets secrete insulin, which reduces the level.